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Mid-Program Milestone Checkpoint

Problem

The **Permission & Autonomous Control** granted to AI systems during system development carries inherent risks. Therefore, the importance of classifying, verifying the rationale behind each autonomous decision an AI makes throughout the SDLC, and predicting the outcomes of each decision increases with the scale and criticality of the system being developed. Consequences can include direct and indirect losses from agentic errors, the hidden cost of oversight, and potential adoption delays. As AI agents become increasingly capable of taking actions across multiple platforms, security processes (which are heavily dependent on traditional human-focused decision-making) are not keeping pace. As a result, it can be difficult for an organization to determine whether an autonomous action is appropriate before consequences arise.

User

Businesses exploring modern/cutting-edge AI integration and Developers. As AI technology advances, software and hardware adaptations for maximum efficiency (energy) and cost-effectiveness (investment & return) are being developed and will enter the live market. At this point, an engineer, who is overseeing the agent deployment moving from assistance to execution faster than governance practices, legal frameworks, and security tooling are maturing, needs something to cover the widening gap between what agents *can* do and what organizations can *safely allow* them to do.

AI capability

Rather than relying on fixed, human-written approval rules, our system uses AI to interpret the context, intent, and likely impact of each decision as it is made across the SDLC. Based on classifications, the system will recommend or enforce a permission level for each action, then execute automatically, execute with logging, pause for human approval, or block. Human approval is reserved for decisions that truly require it, thereby reducing the hidden cost of oversight without removing control.

Agent

An intelligent system could help users control AI agents' actions by understanding the permissions and possible outcomes of each action, as today's security measures are not keeping pace with AI agents' increasing autonomy. The system can understand the context of AI's actions, such as what it's trying to accomplish and its permissions. Decisions about where AI can act within an organization can be evaluated more carefully if greater emphasis is placed on obtaining its permission. Finally, there can be better coordination between developers and security teams. Using agentic AI, the system can act as a checkpoint, independently evaluating and approving or blocking other agents' actions as they occur, rather than waiting for a human to catch problems afterward.

Knowledge & Tools

AI Safety / AI Alignment	Human-in-the-loop, oversight, keeping agent actions aligned with user intent
AI Guardrails / Runtime Guardrails	Classifying and blocking agent actions as they happen
Computer Security (InfoSec)	Least privilege, access control, sandboxing, threat modeling
Authorization / Access Control	Permission models such as RBAC, capability-based security, and policy engines
Risk Assessment / Risk Scoring	Safe/Neutral/Warning labels, reversibility, blast radius
LLM Security	Prompt-injection defense, LLM-as-judge, tool-use safety
AI Governance	Audit logs, accountability, policy enforcement

Workflow

1. **Parse**: extract structured fields from the request, such as tool, command, target paths, network destinations, and data involved.
2. **Classify**: assign a permission category.
3. **Predict**: estimate what happens if the request is granted, including side effects, reversibility, and blast radius.
4. **Label**: assign Safe, Neutral, or Warning.
5. **Act**: auto-approve, auto-deny, or escalate to the human.
6. **Log**: record the decision and its reasoning for audit and retraining.